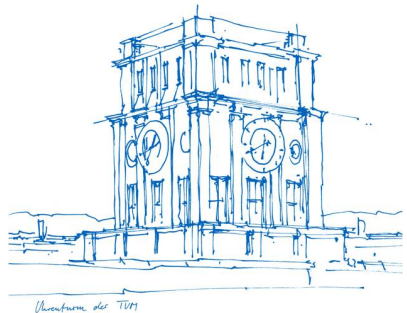


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in
DEM-generated zeolite beds

Technical University of Munich
Chair of Process Systems Engineering

Ali Khalifa & Heiko Briesen



Presentation outline

Motivation and research question

Pore-network reference model

Reduction and diffusivity prediction

Results and robustness

Conclusions and next steps

Section 1 of 5

Motivation and research question

Why should inlet-connected pathways matter?

Why examine access from the inlet?

What porosity misses

- Porosity measures how much void space exists.
- It does not describe bottlenecks, path lengths or parallel connections.
- Similar void fractions can therefore hide different gas-supply pathways.

Why the inlet is the reference

- Gas enters from a reservoir below the bed.
- Adsorption removes gas as it diffuses into the packing.
- Access from the inlet may therefore help explain transient uptake differences.

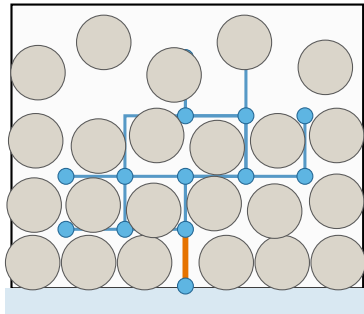
Application: CO₂ adsorption in zeolite 13X, used here as a controlled model problem.

Hypothesis: inlet-connected diffusive pathways contain information that porosity misses.

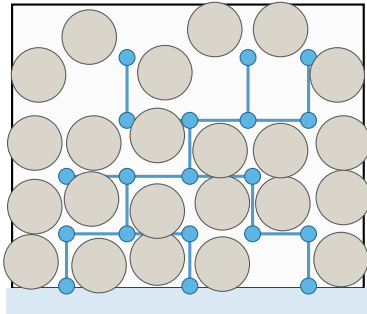
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



● pore body

— diffusive throat

● zeolite bead

— inlet bottleneck

Bed height (20 beds)

$H/d_p = 22.06\text{--}22.41$

Schematic, not to scale

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

Section 2 of 5

Pore-network reference model

From particle geometry to the mean uptake curve

Twenty beds isolate variability caused by random arrangement

Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
inlet c_{CO_2}	6.05 mol m^{-3}
simulation time	5000 s

Controlled diffusion–adsorption problem

- 20 beds generated by LAMMPS DEM: gravity pouring followed by damped settling
- same beads, tube and filling protocol; only the insertion seed changes
- dry, isothermal diffusion–adsorption; gas supplied from below, no imposed flow
- no heat effects or competitive water adsorption

$c_{\text{in}} = y_{\text{CO}_2} p / (RT) \approx 6.05 \text{ mol m}^{-3}$ at 15 vol.%, 1 bar and 298 K.

15% represents coal-combustion flue gas (typical CO_2 : 10–15%).

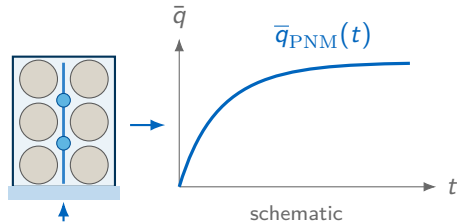
A dry, cooled reference gas fixes the driving concentration across all beds.

Source: U.S. DOE, Quadrennial Technology Review (2015), Sec. 4.E.

The ensemble tests arrangement effects under otherwise identical conditions.

Pore-network model: the uptake reference

A reference curve for each bed



Identical beads:

$$\bar{q}_{\text{PNM}}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t).$$

For each pore:

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - m_b \sum_{k \in \mathcal{B}(i)} w_{ik} \frac{dq_k}{dt}.$$

For each bead:

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

q_k^* : equilibrium target from the mean gas concentration around bead k and the saturation loading q_s (Langmuir isotherm).

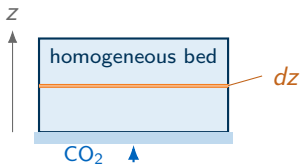
One coupled system: one concentration per pore, one loading per bead.

Section 3 of 5

Reduction and diffusivity prediction

Fit a reduced model, then predict its coefficient

Reduced 1D model and fitting target



In practice: 100 finite-volume cells.

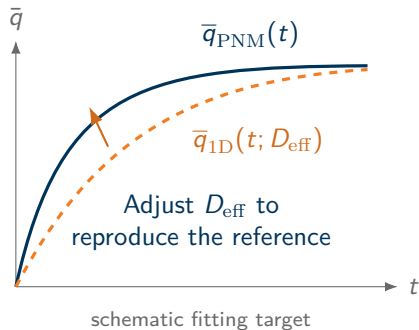
$$\bar{q}_{1D}(t; D_{\text{eff}}) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D_{\text{eff}}).$$

Gas and loading balances in each cell

$$\varepsilon_b \frac{\partial c}{\partial t} = D_{\text{eff}} \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t},$$

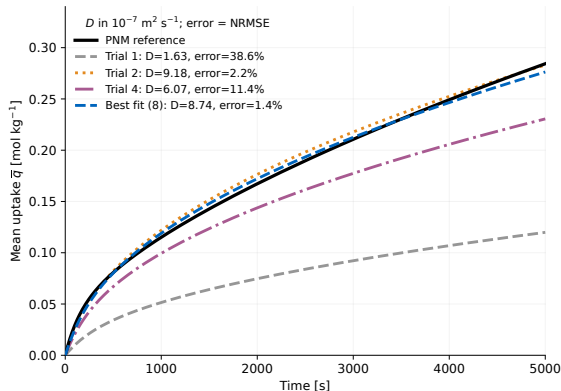
$$\frac{\partial q}{\partial t} = k_{\text{LDF}} (q^* - q).$$

Match the PNM reference



Fit effective diffusivity to the uptake curve

Trial curves: bed 18427



Selected trials; each curve uses one constant diffusivity.

Trial $D \rightarrow$ one transient 1D solve $\rightarrow$ average cell loadings $\rightarrow$ compare

Compare with the PNM reference

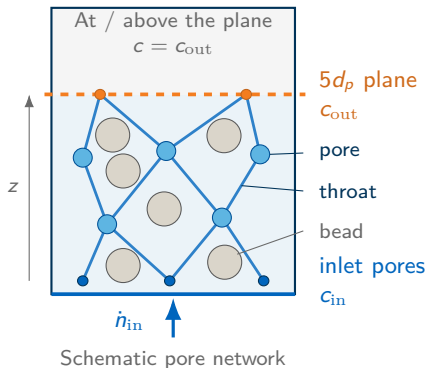
$$e_m(D) = \bar{q}_{1D}(t_m; D) - \bar{q}_{\text{PNM}}(t_m)$$

$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)}$$

w_m : normalized weights accounting for unequal spacing of output times.

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D)$$

Inlet access from a steady diffusion solve



Solve steady pore concentrations

No accumulation, adsorption or imposed flow.

$$0 = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i)$$

↓ Sum throat fluxes leaving the inlet

$$\dot{n}_{in} = \sum_{\substack{i \in \mathcal{I} \\ j \in \mathcal{N}(i) \setminus \mathcal{I}}} G_{ij}(c_{in} - c_j)$$

↓ Divide by the concentration difference

$$G_{access} = \frac{\dot{n}_{in}}{c_{in} - c_{out}} \quad [\text{m}^3/\text{s}]$$

$\mathcal{I}$: inlet-pore set.

G_{access}

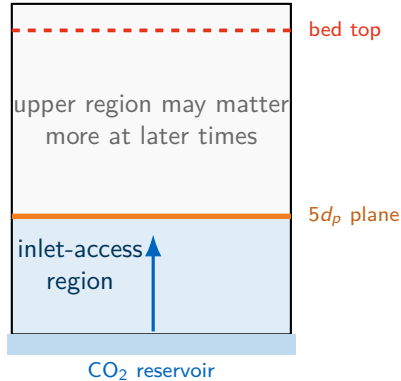


calibrated relation
 $D_{eff} = f(G_{access})$



1D uptake curve

Why use an interior plane?



A plane at $5d_p$ includes

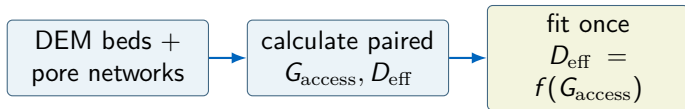
- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

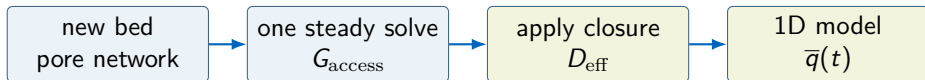
- measure conductance through the entire bed
- include upper bottlenecks that may matter more at later times

Development and later use are two different workflows

Develop closure



Predict for a new bed



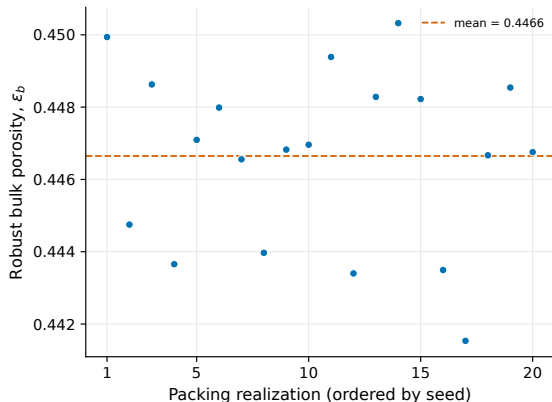
For a new bed, the transient pore-network adsorption model is no longer required.

Section 4 of 5

Results and robustness

Test the physical hypothesis and the model reduction

The twenty beds have nearly equal bulk porosity



Bulk-porosity CV: 0.55%.

Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

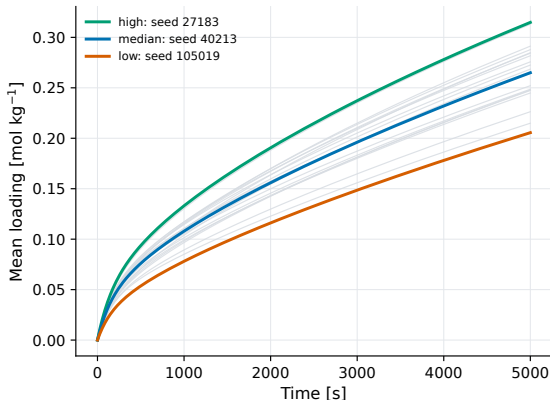
CV: coefficient of variation

$$CV_{\epsilon} = s_{\epsilon} / \bar{\epsilon}$$

s_{ϵ} : standard deviation

$\bar{\epsilon}$: mean porosity

Similar porosity still produces different uptake curves

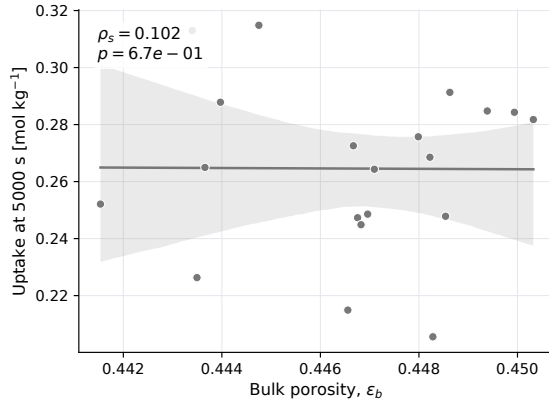


Network reference

- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

At 5000 s: 0.206–0.315 mol kg⁻¹, with an uptake CV of 11.0%.

Bulk porosity alone does not explain final uptake

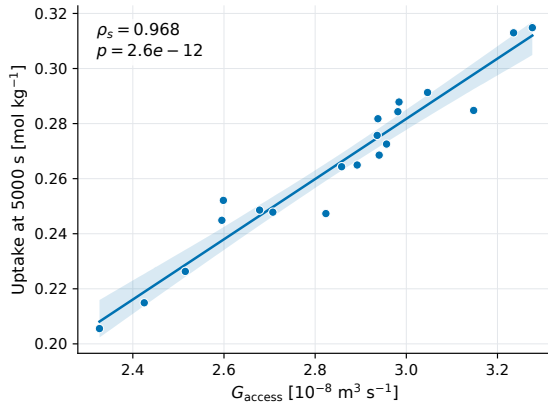


Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

Void volume alone does not describe access to the adsorbent.

Accessibility explains differences in final uptake



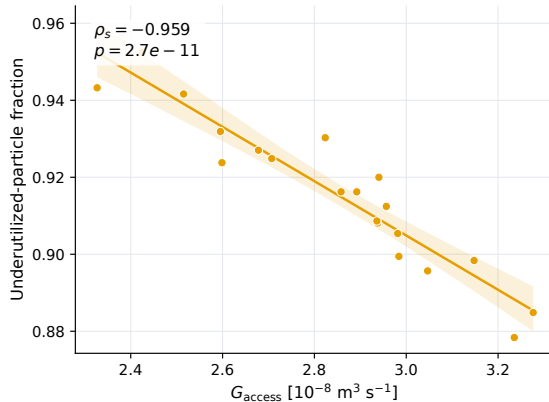
Steady network descriptor

- G_{access} measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

Spearman $\rho_s = 0.968$.

Inlet-connected pathways explain variability that porosity misses.

Greater accessibility reduces particle underutilization



Higher G_{access} accompanies fewer underutilized particles.

Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

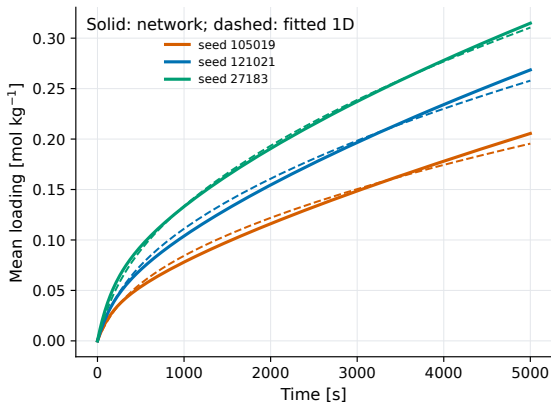
$I_k = 1$ when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

$t_f = 5000$ s.

The indicator equals one for a particle below half the inlet-equilibrium loading.

One fitted D_{eff} reproduces each uptake curve



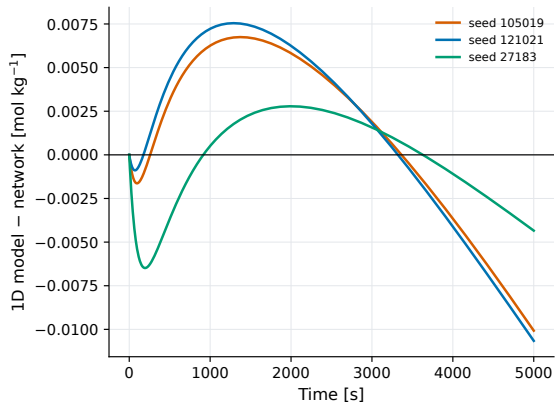
Per-bed optimization

1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

Solid: network.
Dashed: fitted 1D model.

Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

Residuals show the remaining reduction error



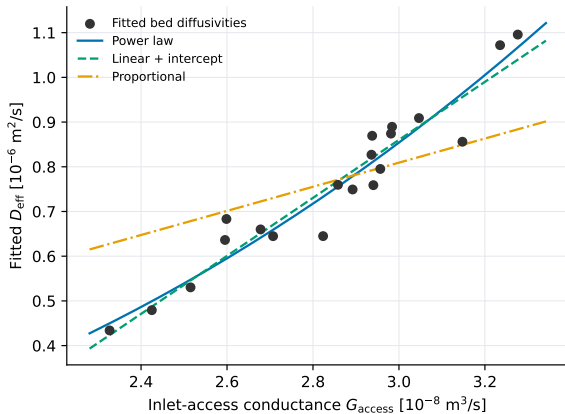
Curve residual

$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

- positive: excess predicted uptake
- negative: insufficient predicted uptake

Time-weighted NRMSE across all beds: 0.84–2.46%.

Three relationships use the same inlet conductance



Predict D_{eff} from G_{access}

Proportional: $D_{\text{eff}} = \alpha_G G_{\text{access}}$

Linear: $D_{\text{eff}} = a_G G_{\text{access}} + b_G$

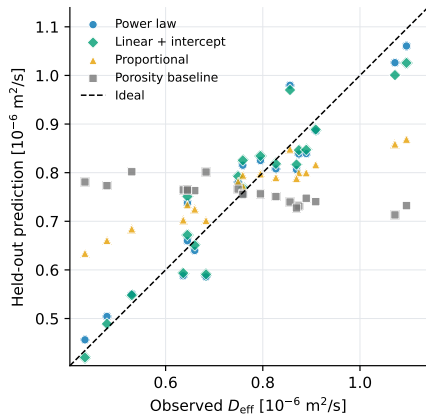
Power law: $D_{\text{eff}} = C_G G_{\text{access}}^n$

All coefficients are calibrated.
Their units make D_{eff} a diffusivity.

The power-law exponent is
 $n \approx 2.52$ for these twenty beds.

The predictor stays fixed; the comparison tests the relationship's form.

Conductance predicts diffusivity for an omitted bed



Leave-one-bed-out comparison

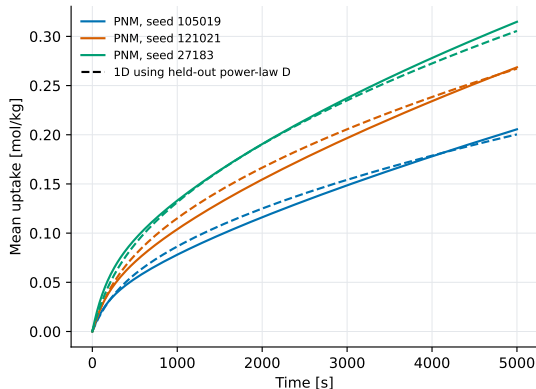
Predictor / form	R^2
G_{access} / power law	0.906
G_{access} / linear	0.898
G_{access} / proportional	0.558
Porosity / linear	-0.234

Fit every coefficient on nineteen beds; predict the omitted bed.

Repeat for all twenty beds.

Power-law and linear predictions are close; both outperform porosity in this ensemble.

Predicted diffusivity reproduces the uptake history



Dashed: power law fitted without the plotted seed.

Mean NRMSE (20 beds)

Relationship	Error
G_{access} / power law	2.61%

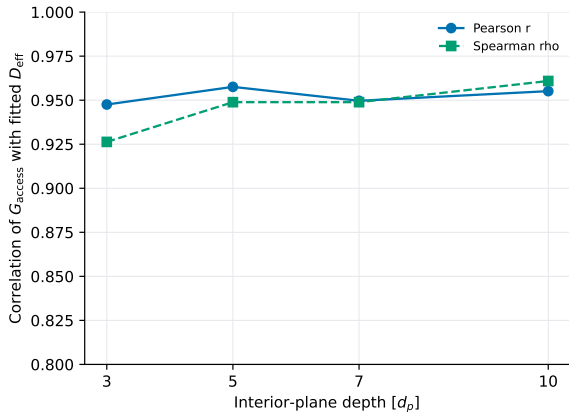
For each omitted bed

1. Fit the power law on the other 19 beds.
2. Predict this bed's D_{eff} from its G_{access} .
3. Run the 1D model without fitting its uptake curve.

One predicted D_{eff} per bed, constant in time.

Leave one bed out, predict its diffusivity, then predict its uptake history.

Conductance correlation across diagnostic depths



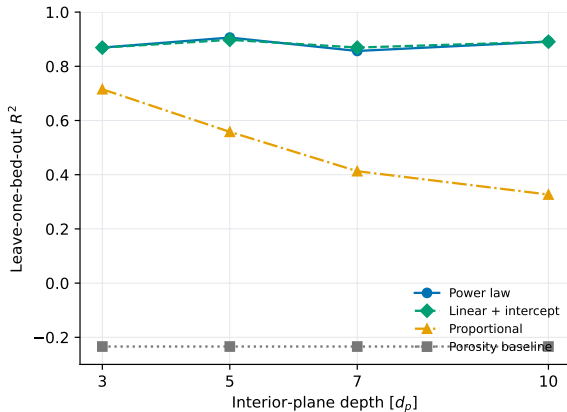
Depth sensitivity

- calculate G_{access} to planes at 3, 5, 7, $10d_p$
- Pearson: linear association
- Spearman: rank association

Depth is measured above the network's lowest pore.

The conductance–diffusivity association persists across the tested interior planes.

Conductance-model performance across diagnostic depths



Four curves at every depth

$$D_{\text{eff}} = C_G G_{\text{access}}^n$$

$$D_{\text{eff}} = a_G G_{\text{access}} + b_G$$

$$D_{\text{eff}} = \alpha_G G_{\text{access}}$$

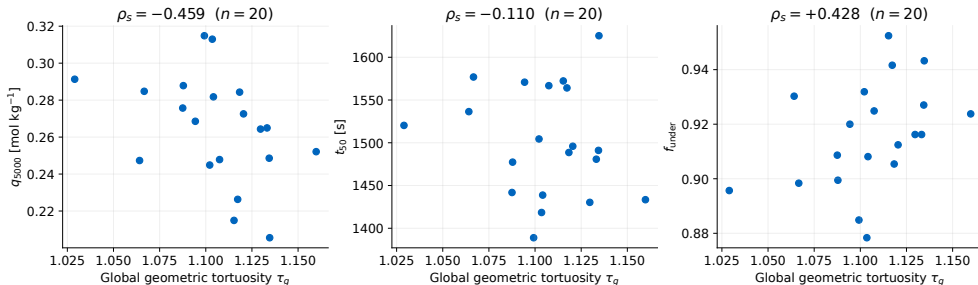
$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b$$

Refit on nineteen beds at each depth; predict the omitted bed.

The porosity baseline is independent of plane depth.

Power-law and linear conductance models remain predictive across the tested depths.

Global tortuosity has weak to moderate associations



- τ_g : shortest inlet–outlet path divided by the axial boundary span.
- t_{50} refers to half of each bed's uptake at 5000 s, not equilibrium.
- Original G_{access} : $\rho_s(q_{5000}) = 0.968$, $\rho_s(f_{\text{under}}) = -0.959$.

This geometric tortuosity descriptor does not reproduce the strong uptake association.

Inlet depth and translated slabs test different questions

Existing inlet-depth test

$$G(0 \rightarrow 3d_p), \quad G(0 \rightarrow 5d_p)$$

$$G(0 \rightarrow 7d_p), \quad G(0 \rightarrow 10d_p)$$

Retain the hull-labelled inlet pores. Move the interior sink plane.

Tests sensitivity to diagnostic depth.

Inside each slab: $c = 1$ in the bottom $1d_p$, $c = 0$ in the top $1d_p$. Remove exterior links. Check 0.5 and $1.5d_p$ bands.

New translated-window control

$$G_{0-5}, \quad G_{5-10}, \quad G_{10-15}, \quad G_{15-20}$$

Translate a fixed $5d_p$ window. Use identical boundary bands at every position.

Tests whether local slab conductance predicts the same outcomes.

All 20 networks evaluated. The 20–25 d_p window exceeds every bed.

Why the two $0-5d_p$ diagnostics differ

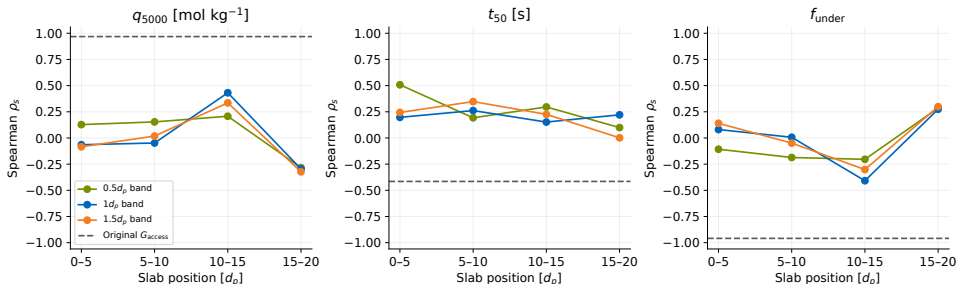
The old source is the labelled inlet. The new source is a full pore band.

Source held at $c = 1$	Sink held at $c = 0$	Uptake ρ_s
Original inlet pores	All pores above $5d_p$	+0.968
Original inlet pores	All pores above $4d_p$	+0.968
All pores in bottom $1d_p$	All pores above $5d_p$	+0.023
All pores in bottom $1d_p$	All pores above $4d_p$	-0.065

Same 20 networks and uptake outcomes. Each row changes only the steady diagnostic.

Changing the source-pore selection removes the strong association. These are different boundary tests.

Translated slabs do not reproduce the uptake association



- With 1 d_p bands, uptake $\rho_s = -0.065, -0.048, +0.432, -0.296$.
- Even the matched 0-5 d_p slab has a weak uptake association.
- Original inlet pores and full source bands impose different boundary conditions.

The inlet boundary definition remains a competing explanation for the strong G_{access} result.

Section 5 of 5

Conclusions and next steps

What is established and what remains to validate

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected conductance strongly predicts utilization
- one D_{eff} reproduces each detailed curve accurately
- G_{access} provides an interpretable predictive closure

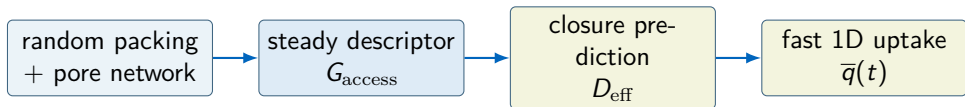
Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- inlet-specific causation or a universal closure

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



20 beds
controlled ensemble

1.58%
mean reduction error

2.61%
held-out curve error

Thank you!

Any questions?

Appendix A

Pore-network model

Coupled balances, equilibrium and reference uptake

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its valid circumcentre represents a pore node
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels
- 2^{22} Sobol points sample the solid and gas space.
- Gas-point counts allocate the calculated void volume to pores.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$D_m = 1.5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$: molecular CO_2 diffusivity in the gas.

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{ s}^{-1}$.

The pore network solves diffusion and adsorption together

Gas balance for each pore i

$$V_i \frac{dc_i}{dt} = \sum_{j \in \mathcal{N}(i)} G_{ij}(c_j - c_i) + G_{iR}(c_R - c_i) - S_i.$$

- V_i : gas volume, c_i : gas concentration
- G_{ij} : throat conductance [$\text{m}^3 \text{s}^{-1}$]
- G_{iR} : connection to reservoir at c_R , zero away from the inlet
- S_i : net gas transfer to adjacent beads [mol s^{-1}]

Uptake for each bead k

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

- q_k : current loading [mol kg^{-1}]
- k_{LDF} [s^{-1}]: linear-driving-force rate coefficient
- $1/k_{\text{LDF}}$: relaxation time at fixed gas concentration

One coupled system: one concentration per pore and one loading per bead.

Local equilibrium sets the adsorption target

Local equilibrium loading

$$p_k = c_k R T, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

- c_k : average gas concentration around bead k
- p_k : local CO₂ partial pressure [Pa]
- q_k^* : equilibrium target [mol kg⁻¹]

Fixed material parameters

- q_s : maximum CO₂ loading [mol kg⁻¹]
- b : Langmuir affinity [Pa⁻¹], so $b p_k$ is dimensionless
- k_{LDF} : kinetic coefficient [s⁻¹]

Actual loading evolves in time

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

Local gas concentrations set the target. Kinetics determine how quickly beads approach it.

Particle uptake: the prescribed LDF coefficient

Literature input and conversion

$$\frac{D_{\text{particle}}}{R_p^2} = 5.7 \times 10^{-4} \text{ s}^{-1}$$

Commercial 13X beads; CO₂ in N₂,
38°C and 0.1 bar (Hu et al.).

Spherical-particle LDF approximation:

$$k_{\text{LDF}} \approx 15 \frac{D_{\text{particle}}}{R_p^2}$$

$$k_{\text{LDF}} = 8.55 \times 10^{-3} \text{ s}^{-1}$$

Assumption: the 38°C literature value is used unchanged at 298.15 K (25°C). The factor-15 conversion is ours, not a directly reported LDF coefficient.

Source: Hu et al., Adsorption 20 (2014), 121–135. doi:10.1007/s10450-013-9554-z. Parameter provenance: project reproducibility notes, Sec. 9.5.

Meaning in our model

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k)$$

- One prescribed coefficient in both PNM and 1D models.
- Represents uptake within a bead; distinct from bed-scale D_{eff} .
- At a fixed equilibrium target: relaxation time $1/k_{\text{LDF}} \approx 117 \text{ s}$.

The pore network delivers the reference uptake curve

Resolved transient outputs

- Gas concentration $c_i(t)$ in every pore
- Adsorbed loading $q_k(t)$ of every bead
- Different bead loadings reveal incomplete utilization

The detailed model retains the actual pore connections and bead associations.

Bed-average uptake

$$\bar{q}_{\text{PNM}}(t) = \frac{\sum_k m_k q_k(t)}{\sum_k m_k}.$$

For identical beads:

$$\bar{q}_{\text{PNM}}(t) = \frac{1}{N_b} \sum_{k=1}^{N_b} q_k(t).$$

This curve is the reference for fitting the reduced 1D model.

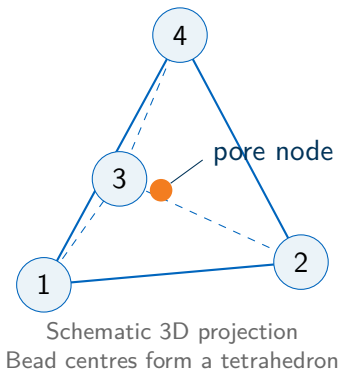
Next: can a homogeneous 1D bed reproduce this complete uptake curve?

Appendix B

How the pore network is constructed

Particle geometry, Sobol volume allocation and transport

1. Particle positions define candidate pore nodes

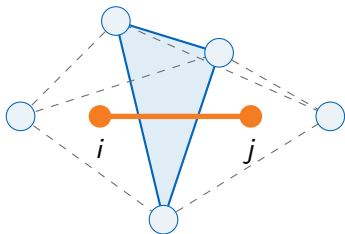


- Start with known bead centres, radii and bed boundaries.
- Delaunay tetrahedralization groups neighbouring centres into sets of four.
- Locate the circumcentre: the point equidistant from those four centres.
- Retain valid points inside the bed and outside the solid as pore nodes.

A node represents a gas region. Its volume is assigned later.

The particle geometry determines where the pore nodes are placed.

2. Shared particle faces define pore connections



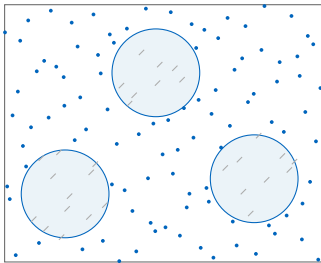
Two tetrahedra share three beads
Schematic 3D projection

- Two neighbouring tetrahedra share a triangular face.
- The three beads around that face define an opening.
- A valid opening connects pore nodes i and j through a throat.
- Estimate throat length L_{ij} and circular area A_{ij} from geometry.

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}.$$

At this stage we know the nodes and connections. Sobol sampling has not been used.

3. Sobol points measure the gas space



Illustrative 2D Sobol sample

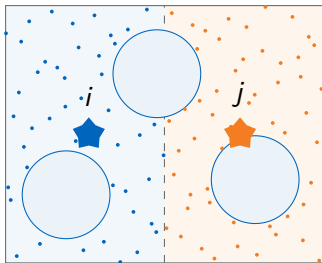
Blue: gas points; grey crosses: solid points

- Distribute $2^{22} \approx 4.2$ million sample points through the cylindrical bed.
- Sobol sampling covers the domain with few gaps and clusters.
- Check each point against the bead surfaces.
- Discard points inside solid particles and retain gas-space points.

These are measuring points, not extra pores or mesh cells.

Sobol sampling estimates volume. It does not generate network connections.

4. Assign gas-space samples to the nearest pore



Two illustrative pore nodes
Each gas point takes its nearest node's colour

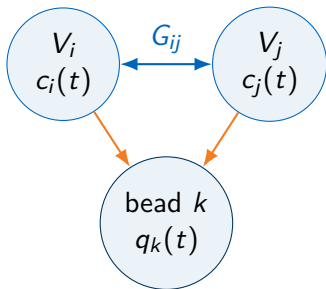
- Assign every retained gas point to its nearest pore node.
- Count the assigned points: N_i for pore i .
- A larger count gives a larger share of the calculated void volume.

$$V_i = V_{\text{void}} \frac{N_i + 0.5}{\sum_j (N_j + 0.5)}.$$

- The small 0.5 correction prevents zero volumes.
- Normalization ensures $\sum_i V_i = V_{\text{void}}$.

The total volume is conserved, while each individual pore volume is an estimate.

5. The network carries coupled diffusion and adsorption



Schematic during net adsorption

- Pore volumes store gas: $n_i = V_i c_i$.
- Throats transfer gas according to concentration differences.
- Associated beads adsorb gas from adjacent pores.
- Boundary connections supply gas from the inlet reservoir.

$$\dot{n}_{j \rightarrow i} = G_{ij}(c_j - c_i), \quad \frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k).$$

Throats control gas movement. Pore volumes control local gas storage.

Appendix C

Homogeneous 1D model

Discretization, balances and bed-average uptake

Reduce the detailed network to a homogeneous 1D bed

Detailed pore-network model

- Actual pores and throats from each packing
- One gas concentration per pore
- One loading per bead
- Reference mean uptake $\bar{q}_{\text{PNM}}(t)$

Homogeneous 1D model

- Replace the network by 100 equal axial slices
- Each slice spans the full tube cross-section
- One gas concentration and one mean solid loading per slice
- Represent unresolved pathways with a single diffusivity D

Fit D so the reduced model reproduces the network's mean uptake history.

The homogeneous 1D uptake model

Gas balance in each axial cell

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}.$$

Adsorbent loading in each axial cell

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} \left[\frac{q_s b R T c}{1 + b R T c} - q \right].$$

- Divide the bed height into 100 equal slices, each spanning the tube cross-section.
- Solve gas concentration and mean solid loading together in each slice.
- Trial D changes c , which changes adsorption.

ε_b : **bed porosity, uniform in all 100 cells of each bed.**

Average the calculated loading

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

For 100 equal axial cells:

$$\bar{q}_{1D}(t; D) = \frac{1}{100} \sum_{\ell=1}^{100} q_{\ell}(t; D).$$

- Match bed height, porosity and inlet condition to the network.
- Keep isotherm, kinetics and initial state fixed. Only D changes.